选择：

1. LC 电路频率 f，最大电流 I,电路的能量扩大二倍，电容的最大电压扩大多少倍
2. 地球的磁偶极矩可以看作赤道上的电荷绕轴自转产生，估算赤道上的电荷（没有数据）
3. 正交偏振片中间加一个 45° 偏振片，光强变为多少
4. 带均匀电荷圆形气球体积减少，U 怎么变
5. 2. 单缝衍射，红光的第二极大和紫光的第一极大在空间上重复了，如果减小缝宽，这两个区域会相互分离吗，如果分离的话两区域是如何移动的。

填空：

1. 默写麦克斯韦方程组
2. 默写一维含时薛定谔方程，动量算符，能量算符
3. 默写电磁波五条性质
4. **25.** If the wavelength of the incident light is $\lambda = 572$ nm, rays A and B in Fig. 41-24 are out of phase by 1.50λ. Find the thickness d of the film.

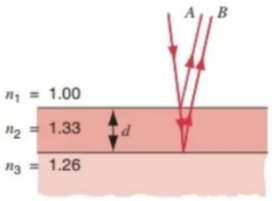


FIGURE 41-24. Exercise 25.

5. 距信号发射塔 R 的飞机接收到发射塔发出的波的能流密度 w，求 Emax，Bmax，信号发射塔发出的功率
6. 一排电流垂直纸面向里，宽度 w，电流大小 I,求据一端 L 处的 B 的大小和方向
7. 一个薄透镜放入水中（n=4/3），薄透镜内部是空气，左侧为 r=2 的圆弧，右侧为直线，如下图。求焦距

5. Show that the focal length f for a thin lens whose index of refraction is n and that is immersed in a fluid whose index of refraction is n' is given by

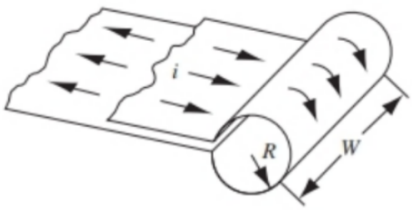
$$\frac{1}{f} = \frac{n - n'}{n'}\left(\frac{1}{r_1} - \frac{1}{r_2}\right).$$

第 7 题对应的作业原题

8. 一个锯子和一个斧子（忘记是什么工具了不过不重要）并排放置，给 Q,U 求 C,改变 Q 求变换之后的 U

大题：

2. A wide copper strip of width W is bent into a piece of slender tubing of radius R with two plane extensions, as shown in Fig. 36-24. A current i flows through the strip, distributed uniformly over its width. In this way a "one-turn solenoid" has been formed. (*a*) Derive an expression for the magnitude of the magnetic field $\vec{\mathbf{B}}$ in the tubular part (far away from the edges). (Hint: Assume that the field outside this one-turn solenoid is negligibly small.) (*b*) Find also the inductance of this one-turn solenoid, neglecting the two plane extensions.



1. **FIGURE 36-24.** Problem 2.

14. Figure 38-30 shows a parallel-plate capacitor being charged. (*a*) Show that the Poynting vector $\vec{\mathbf{S}}$ points everywhere radially into the cylindrical volume. (*b*) Show that the rate at which energy flows into this volume, calculated by integrating the Poynting vector over the cylindrical boundary of this volume, is equal to the rate at which the stored electrostatic energy increases; that is,

$$\int \vec{\mathbf{S}} \cdot d\vec{\mathbf{A}} = Ad \frac{d}{dt}\left(\frac{1}{2}\epsilon_0 E^2\right),$$

where Ad is the volume of the capacitor and $\frac{1}{2}\epsilon_0 E^2$ is the energy density for all points within that volume. This analysis shows that, according to the Poynting vector point of view, the energy stored in a capacitor does not enter it through the wires but through the space around the wires and the plates. (Hint: To find $\vec{\mathbf{S}}$ we must first find $\vec{\mathbf{B}}$, which is the magnetic field set up by the displacement current during the charging process; see Fig. 38-2. Ignore fringing of the lines of $\vec{\mathbf{E}}$.)

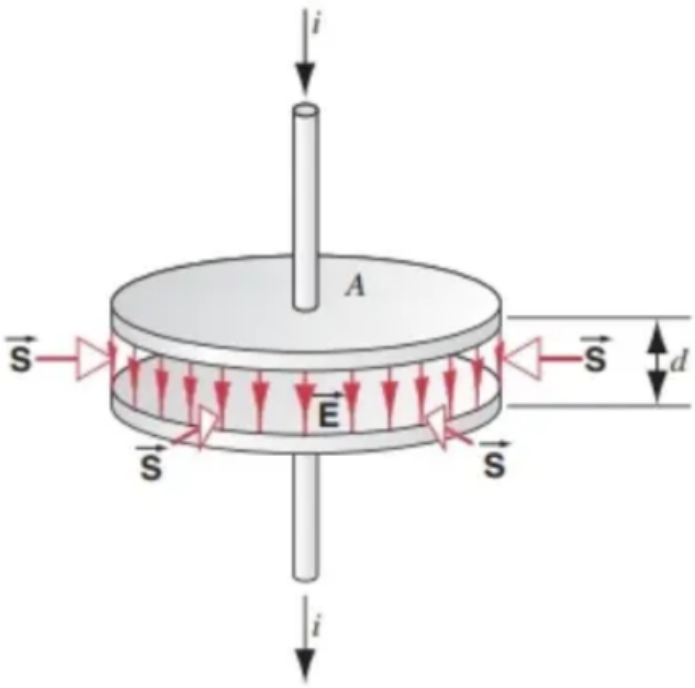


FIGURE 38-30. Problem 14.

2.

3. 衍射光栅，两束光波长分别为 589nm 和 589.59nm，知道 d

（1）第一主极大的角度（2）第一主极大角度之差（3）最小分辨率（4）想要分辨这两束光，

光栅数需满足什么要求

4. 附加题

一个无限长线圈，知道单位长度的 n，半径 R 和电流 I,有两个共轴圆筒半径分别为 a,b,高度为均为 l（两圆筒在同一高度）l>>b>R>a。内部圆筒带电荷+Q，外部圆筒带电荷-Q 电流逐渐减小到 0，求两圆筒角动量。